

CHAPTER I

INTRODUCTION

1.1. Background

With more than 60 years since the first computer created, the world are already into an exciting era of Information. Since the first ENIAC created at Massachusetts Institute of Technology more and more hard labor are being taken by the machines. Many Factory owner started to use Robots instead of human because of its efficiency and precision. Computers replace typing machines. Suddenly, we see computers everywhere from those big “Supercomputers” to those little Raspberry Pi. From designing an airplane to creating a thesis. Everywhere computers assisting humans in their job. It's been clear that Human's Civilization is at a dawn of new era, The Information Era.

The development of computers affects many things in human life. This development ushered humanity into a new era, an Information era where information spread and can be accessible by many. One of the technology that affected heavily by the advancement of computers technology is telecommunication technology. Nowadays we can easily find smart phones, a phone that have embedded computers making them able not only as a standard phones but also able to become a general-purpose device that can be programmed to carry out a set of arithmetic or logical operations automatically. Since a sequence of operations can be readily changed, it can solve more than one kind of problem.

The introduction of smart phone lead to a boom of mobile applications developed specially for smart phones. The human life are becoming easier for each passing day.

But, just like the two sides of a coin. There are always bad things following good things. As computers getting our life easier each day, it becomes the new magnet that promote “couch potato”.

Couch potato is a term that used for people that are too fixated watching television and resulted that they started getting weight and end up as an obese. The term couch is refer to the habit of the person who like to sit on a couch all day. Potato refer to potato chips a favorite snacks for many people when watching TV. So couch potato is a term that refer to people that like to watch TV while eating potato chips. This habit make the person becoming overweight and can lead to become obese.

Obesity is a medical condition in which excess body fat has accumulated to the extent that it may have a negative effect on health, leading to reduced life expectancy and/or increased health problems. People are considered obese when their body mass index (BMI), measurement obtained by dividing a person's weight by the square of the person's height, exceeds 30kg/m^2 , with the range $25\text{-}30\text{ kg/m}^2$ defined as overweight. [25]

Obesity increases the likelihood of various diseases, particularly heart disease, type 2 diabetes, obstructive sleep apnea, certain types of cancer, and osteoarthritis. Obesity is most commonly caused by a combination of excessive food energy intake, lack of physical activity, and genetic susceptibility, although a

few cases are caused primarily by genes, endocrine disorders, medications, or psychiatric illness. Evidence to support the view that some obese people eat little yet gain weight due to a slow metabolism is limited. On average, obese people have a greater energy expenditure than their thin counterparts due to the energy required to maintain an increased body mass.

Despite its growing number in many first world country most notably United States of America there are still little data exist regarding the use of mobile health approaches in the management of adolescent obesity.

1.2.Problem Description

People with a mild eating disorders become the highly targeted consumer to taste and feel the development of applications in this modern era. In this growing age, there are a lot of diet application did not educate them well, even making people do a harmful diet regime that can hurt both their mind and body. Thus, an application that could help the user and also learn from the application to have a healthy lifestyle.

1.3.Thesis Objective

The objective of this thesis are to develop a Diet app in such that it emphasis more to the healthy lifestyle instead of just plain weight loss. Currently there are not many diet application that promote healthy lifestyle. This thesis is expected to become an application that not only able to achieve its target but also able to sustain it for a longer time.

This thesis is developed under the intent to be able to do these things:

- Assists the user to know the calorie amount of foods.

- Record the amount of calorie they eat every day.
- Count Body Metabolism Rate and Daily Calorie Needs.
- Help users reach their desired weight through healthy living habit.

This thesis will be developed to Android phone and written using Java programming language as the main programming language. The tools used are Android Studios as main IDE platform. The application will be named “Diet Project”.

1.4.Scope and limitations

This thesis is expected to achieve the goals and objectives. Therefore this research focuses on the following ability:

- Diet Project is a diet application for mobile and thus is specified for one user only which can be run on mobile devices with minimal Android OS version 4.0 or higher installed.
- Diet Project is should be able to count Body Metabolism Rate, and daily calorie target according to the target the users have.
- Diet Project has some usuall menu such as options, add activity, and add item.
- Diet Project is developed using Android Studio to run on Android mobile platform.

Diet Project focuses on the developing and building functionalities of the application.

To ensure the application can stay on target and not becoming too wide and thus too complicated to be finished with the limited time given. The application will have several limitation put in it:

- Diet Project is not aimed to those people that have a severe eating disorders. i.e.: Morbidly obese or acute anorexia.
- Diet Project can't give suggestion on what the user should eat. It only give user Daily Calorie Target to remind the user of their optimal calorie intake.
- Diet Project is using the diet plan of Calorie restriction diet scheme.

With those in mind the development of Diet Project will be focusses on how to get an application with those scope and limitations.

1.5.Methodology

This research is developed using Rapid Application Development (RAD). Rapid application development (RAD) is both a general term used to refer to alternatives to the conventional waterfall model of software development as well as the name for James Martin's approach to rapid development. In general, RAD approaches to software development put less emphasis on planning tasks and more emphasis on development. In contrast to the waterfall model, which emphasizes rigorous specification and planning, RAD approaches emphasize the necessity of adjusting requirements in reaction to knowledge gained as the project progresses. This causes RAD to use prototypes in addition to or even sometimes in place of design specifications. RAD approaches also emphasize a flexible process that can adapt as the project evolves rather than rigorously defining specifications and plans correctly from the start. In addition to James Martin's RAD methodology, other

approaches to rapid development include agile methods and the spiral model. RAD is especially well suited (although not limited to) developing software that is driven by user interface requirements. Graphical user interface builders are often called rapid application development tools. RAD is a popular software development methodology employing various tools and techniques to quickly produce minimally coded software applications. RAD's essence is prototyping – creating predefined components, structures and methods to quickly develop software models.

In the Rapid Application Development, the development process starts with a quick analysis and design. It defines the preliminary data model using structured techniques. After that, the development process falls to the cycle of prototyping. The high level and detailed design requirements are verified using prototyping. These stages are repeated iteratively until the system is built. It then goes through the next stage which system testing and implementation. Figure 1 shows the Rapid Application Development cycle.

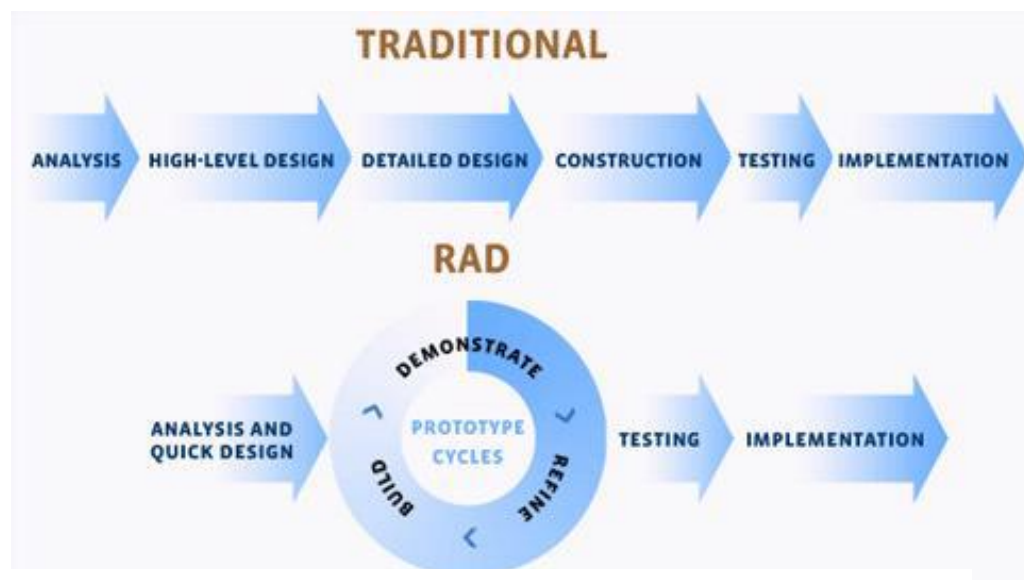


Figure 1.1 Difference between Traditional And RAD application life cycle

(http://www.sanketj.com/2013_04_01_archive.html)

RAD methodology is chosen due to its effectiveness in time due to developing and testing the application at almost the same time.

Based on RAD methodology, the application is tested after each phase of development yield a result (called prototype). If the test result is satisfactory, the development will be carried to the next step.

1.6.Thesis Outline

The Thesis Outline is described as follows:

1.1.1. CHAPTER I INTRODUCTION

This chapter consists of the information of Thesis Background, Problem Statement, Research Objectives, Scope and Limitation, Methodology, and Research Outline.

1.1.2. CHAPTER II LITERATURE REVIEW

This chapter contains the theoretical knowledge in making of thesis. Besides, there is also some comparison work that tries to accomplish work that is similar to the one created before.

1.1.3. CHAPTER III SYSTEM ANALYSIS

This chapter explains the system requirements, features, and a bit of design architecture of the system of the given requirements. The analysis will be explained more detail using UML Use Case Diagram and Swim lane diagram.

1.1.4. CHAPTER IV SYSTEM DESIGN

This chapter explains about the design of the application, includes about its architecture, data, interfaces, and components for the system in a specified requirements.

1.1.5. CHAPTER V SYSTEM DEVELOPMENT

This chapter contains the development and coding of the system. This part is described relatively to each prototype and summarized to converge it to the final working system.

1.1.6. CHAPTER VI SYSTEM TESTING

This chapter evaluates and tests the system that has been made to ensure there are no errors in the system so the entire system works perfectly and the project meet the requirement that decided in the previous phase.

1.1.7. CHAPTER VII CONCLUSIONS AND FUTURE WORK

This chapter includes the conclusions obtained from the result in testing the application, the conclusion of this thesis and possible future works.